



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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REVISION EXAMINATION - 2025

Time Allowed : 3.00 Hours]

CHEMISTRY
PART - I

[Max. Marks : 70

I. Answer the following:**15x1=15**

- The equivalent mass of a trivalent metal element is 9 g eq⁻¹ the molar mass of its anhydrous oxide is
a) 102g b) 27g c) 270g d) 78g
- Assertion** : The spectrum of He⁺ is expected to be similar to that of Halogen
Reason : He⁺ is also one electron system.
a) If both assertion and reason is true and reason is the correct explanation of assertion
b) If both assertion and reason are true but reason is not the correct explanation of assertion
c) If both assertion is true but reason is false d) If both assertion and reason are false
- The element with positive electron gain enthalpy
a) Hydrogen b) Sodium c) Argon d) Fluorine
- Zeolite used to soften hardness of water is hydrated:
a) Sodium aluminium silicate b) Calcium aluminium silicate
c) Zinc aluminium silicate d) Lithium aluminium hydride
- Among the following the least thermally stable
a) K₂CO₃ b) Na₂CO₃ c) BaCO₃ d) Li₂CO₃
- Maximum deviation from ideal gas is expected from
a) CH_{4(g)} b) NH_{3(g)} c) H_{2(g)} d) N_{2(g)}
- The temperature of the system decreases in an -----
a) Isothermal expansion b) Isothermal compression
c) Adiabatic expansion d) adiabatic compression
- Solubility of Carbondioxide gas in cold water can be increased by
a) Increase in pressure b) Decrease in pressure
c) Increase in volume d) None of these
- What is the molality of a 10% W/W aqueous sodium hydroxide solution?
a) 2.778 b) 2.5 c) 10 d) 0.4
- XeF₂ is isostructural with
a) SbCl₂ b) BaCl₂ c) TeF₂ d) ICl₂
- Ortho and para - nitrophenol can be separated by
a) Azeotropic distillation b) Destructive Distillation c) Steam Distillation d) Cannot be separated
- Heterolytic fission of C-C bond results in the Formation of
a) Free Radical b) Carbanion
c) Carbocation d) Carbanion and Carbocation
- Which of the following is aliphatic saturated hydrocarbon
a) C₈H₁₆ b) C₉H₁₈ c) C₈H₁₄ d) All of these
- Chloroform reacts with nitric acid to produce
a) Nitrotoluene b) Nitroglycerine c) Chloropicrin d) Chloropicric acid
- Release of oxides of nitrogen and hydrocarbons into the atmosphere by motor vehicles is prevented by using
a) Grit Chamber b) Scrubbers c) Trickling filters d) Catalytic convertors

PART - II**II. Answer any six questions. Question No. 24 is compulsory.****6x2=12**

- Define relative atomic mass?
- State Aufbau principle?
- Why sodium hydroxide is much more water soluble than sodium chloride?

19. Give the mathematical expression that relates gas volume and moles?
20. State the Third law of thermodynamics.
21. What is Polar covalent bond?
22. What happens when ethylene is passed through cold dilute alkaline potassium permanganate?
23. Which is Considered to be earth's Protective umbrella. Why?
24. Give the IUPAC names of the following compounds.
- i) $\text{CH}_3 - \underset{\text{CH}}{\underset{|}{\text{CH}}} - \underset{\text{Br}}{\underset{|}{\text{CH}}} - \text{CH}_3$
- ii) $\text{CH}_2 = \text{CH} - \text{CH} = \text{CH}_2$

PART - III

III. Answer any six questions. Question No. 33 is compulsory.

6x3=18

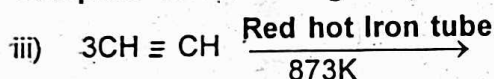
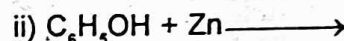
25. Explain the Diagonal relationship?
26. Compare the Structures of H_2O and H_2O_2
27. What is reaction Quotient? Give an example?
28. State and explain Henry's Law?
29. Describe Optical isomerism with suitable example?
30. Write shorts notes on

a) Resonance b) Hyperconjugation

31. What are Freons ? Discuss their uses and environmental effects.?
32. Differentiate BOD and COD?

33. Complete the following. i) 2 - Butyne

Lindlartcatalyst



PART - IV

IV. Answer all the questions.

5x5=25

34. a) Calculate the empirical and molecular formula of a compound containing 76.6% carbon, 6.38% hydrogen and rest oxygen its vapour density is 47. (5)

(OR)

- b) i) How many orbitals are possible for $n=4$? (2)
- ii) Explain the Pauling method for the determination of Ionic radius. (3)

35. a) i) Mention the uses of Deuterium. (2)
- ii) What are isotopes? write the names of Isotopes of Hydrogen. (3)

(OR)

- b) Discuss briefly the similarities between lithium and Megnesium. (5)

36. a) Derive the value of Critical constants in terms of Vander Waals constants. (5)

(OR)

- b) i) Define the Calorific Value of food? What is the Unit of Calorific Value? (2)
- ii) Describe Fajan's rule? (3)

37. a) i) Write a balanced Chemical equation for the equilibrium reaction for which the equilibrium constant is given by Expression. (2)

$$K_c = \frac{[\text{NH}_3]^4 [\text{O}_2]^5}{[\text{N}_2]^4 [\text{H}_2\text{O}]^6}$$

- ii) List the characterstics of Glbb's free energy. (3)

(OR)

- b) i) What is meant by a functional group? Identify the following group in the Following compounds.
- a) acetaldehyde b) oxalic acid c) dlmethyl ether d) methylamine

38. a) Starting from CH_3MgI . How will you prepare the Following.

- a) Acetone (2½) b) Isopropyl Alcohol (2½) (OR)

- b) (i) What is Smog? how does classical Smog differ from Photochemical smog. (3)

- (ii) Write Wurtz fittig Reaction. (2)



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